Math 61 Due: 1/19/24

HW1

0.1 (1) $A \cap B = \{4, 6, 8, 10, 12\}$ all even x where $3 \leq x \leq 13$.

(2) $A \cup B = \{x \in \mathbb{Z}_{>0} \mid 3 \leq x \leq 13 \text{ or } x \text{ is even}\}$

(3) $B \cap C = \{\}$ or $\emptyset$ disjoint

(4) $B \cup C = \{x \in \mathbb{Z}_{>0}\}$

0.2 $A = \{1, 2\}$ $B = \{\{1, 2\}, 1, 2\}$

$A \in B$

$A \subset B$

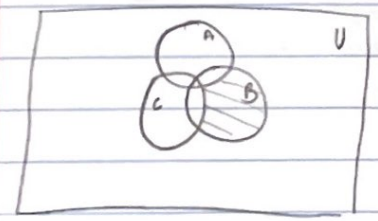
0.3 $A = \{1, 2, 3, 4\}$ $B = \{n \mid n \in A \text{ and } n+m = 8 \text{ for some } m \in C\}$

$C = \{5, 6, 7, 8\}$

$n \mid n \in A$	+	$m \mid m \in C$	=	8	
1	+	7	=	8	$1 \in B$
2	+	6	=	8	$2 \in B$
3	+	5	=	8	$3 \in B$
4	+	?	=	8	

$8 - 4 = 4$. $4 \notin C$. Therefore, $4 \notin B$. Since $4 \in A$, but $4 \notin B$, since $4+m \neq 8$ for any $m \in C$, A is not a subset of B.

0.4



0.5 $f(n) = \lceil n/2 \rceil$ $f: \mathbb{Z} \to \mathbb{Z}$

inj: WTS f is not injective.

$f(1) = \lceil 1/2 \rceil = 1$

$f(2) = \lceil 2/2 \rceil = 1$

$f(1) = f(2)$ but $1 \neq 2$, so $f(n)$ is not injective.

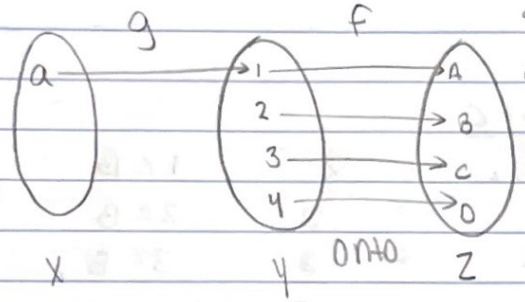
surj: WTS f is surjective.

Let $y \in \mathbb{Z}$, where y is any arbitrary target.

We WTS that for any target $y \in \mathbb{Z}$, there is some possible input $x \in \mathbb{Z}$ where f maps to y. Let $x = 2y$.

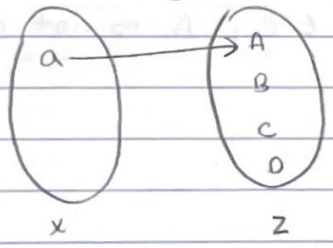
$f(2y) = \lceil 2y/2 \rceil = y$. Since everything in the codomain (all ys) are "hit" by f, f is onto. (all integers)

0.6



Since everything in Z, or the codomain of f, is hit in f, f is onto.

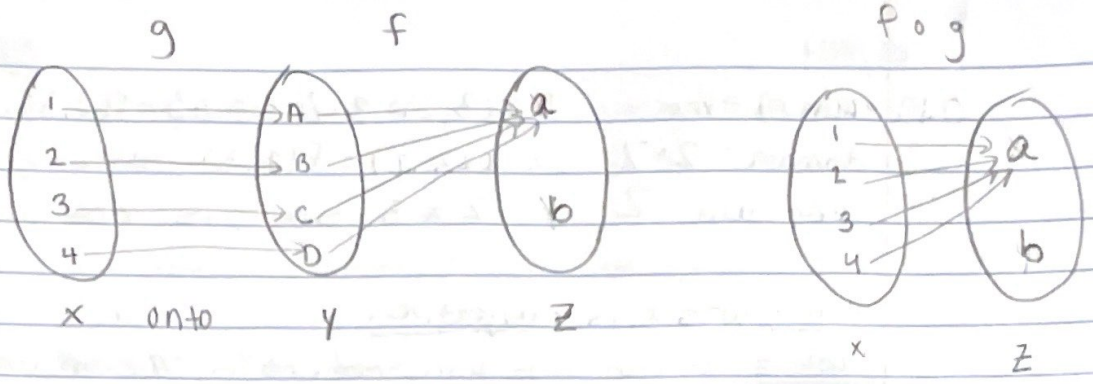
f ∘ g



Since everything in the codomain (right circle) of f ∘ g is not hit, f ∘ g is not onto, or surjective.

NOT ONTO FALSE

0.7



g is onto since everything in its codomain, or y, is hit.
f ∘ g is not onto because not everything in z is hit, such as b. NOT ONTO FALSE

0.8 $f(x) = 4x \bmod 6$

$f(0) = 4(0) \bmod 6 = 0 \bmod 6 = 0$

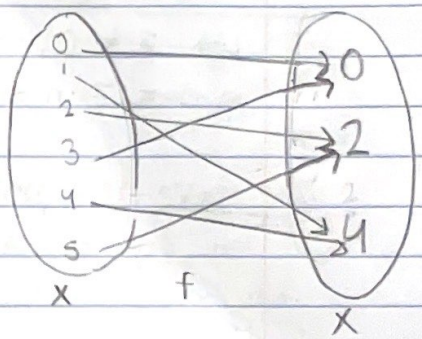
$f(1) = 4(1) \bmod 6 = 4 \bmod 6 = 4$

$f(2) = 4(2) \bmod 6 = 8 \bmod 6 = 2$

$f(3) = 4(3) \bmod 6 = 12 \bmod 6 = 0$

$f(4) = 4(4) \bmod 6 = 16 \bmod 6 = 4$

$f(5) = 4(5) \bmod 6 = 20 \bmod 6 = 2$



$f = \{(0,0), (1,4), (2,2), (3,0), (4,4), (5,2)\}$

f is not one-to-one, but it is onto. For example $f(0) = 0 = f(3)$, so since $f(0) = f(3)$ even though $0 \neq 3$, f is not one-to-one = some numbers in the codomain are hit multiple times. Since every number in the codomain (0,4,2) is hit, the function is onto.

0.9 $f(x) = x \bmod 7$, $x \in \mathbb{Z}$

$f(x) = 3 \quad 3 = x \bmod 7 \Rightarrow x = 7 \cdot q + 3$

$\{x \in \mathbb{Z} \mid f(x) = 3\} = \{7q + 3 \mid q \in \mathbb{Z}\}$

Codomain: $\{0, 1, 2, 3, 4, 5, 6\}$

For any $x \in \mathbb{Z}$, $x \bmod 7$ cannot be greater than 6 or negative, since $x/7$ cannot have a negative remainder or a remainder ≥ 7.

If remainder ≥ 7, you can divide by 7 one more time, leaving a remainder smaller than 7. Therefore, these are all the possible remainders mod 7.

For all integers in $\mathbb{Z}$, there will be at least one where $x \bmod 7 = 0, 1, 2, 3, 4, 5,$ or 6. Therefore, f is surjective with this codomain as all the numbers are hit.

0.10 $f(m,n) = mn$
domain: $\mathbb{Z} \times \mathbb{Z}$
codomain: $\mathbb{Z}$

$f(3,2) = 3 \cdot 2 = 6 = 2 \cdot 3 = f(2,3)$.
$f(3,2) = f(2,3)$ although $3 \neq 2$ and $2 \neq 3$, so f is not one-to-one.

Surj: WTS f is surjective.

Let z be an arbitrary element in $\mathbb{Z}$. We want to show that for any target $z \in \mathbb{Z}$, there's some possible inputs $m, n \in \mathbb{Z}$ where f maps to z.

Let $z = 0$, and $m = z$. For any integer, n,
$m \cdot n = z \cdot n = 0 \cdot n = 0 = z$.

Let $z \neq 0$, and $m = z$. Let $n = 1$. For any integer that z is,
$m \cdot n = z \cdot n = z \cdot 1 = z$.

Since any possible value of z, or the codomain, is hit by f, f is onto.